

REAL ESTATE PORTAL SYSTEM REPORT

Student Name: John Anderson

Course: CIS 344

Date: 05/07/026

INTRODUCTION

The Real Estate Portal System is a web-based application developed using PHP and MySQL. The system allows users to browse, search, and manage property listings efficiently. It supports multiple user roles including agents, buyers, and renters, providing different functionalities based on access level.

This project demonstrates full-stack development by integrating frontend design, backend logic, and relational database management.

OBJECTIVES

The objectives of this project are:

- To design and implement a relational database
- To develop a dynamic web application using PHP
- To perform CRUD operations (Create, Read, Update, Delete)
- To implement user authentication and role-based access
- To integrate advanced SQL features such as views and triggers
- To provide an intuitive user interface for property browsing

SYSTEM FEATURES

User Management

- Secure login system using password_hash() and password_verify()
- User registration functionality
- Session-based authentication
- Role-based access (agent vs user)

Property Management

- Add new properties (agents only)
- Edit existing properties
- Delete properties with constraint handling
- Display all properties dynamically

Search Functionality

Filter properties by city
Filter properties by maximum price

Property Details Page

Displays full property information
Shows property image
Displays agent name and status
Includes inquiry submission form

Favorites System

Users can save properties
Toggle functionality (Save / Remove)
View saved properties

Inquiry System

Users can submit inquiries for properties
Messages stored in database
Linked to both user and property

Image Handling

Images stored in /images folder
Image paths saved in database
Dynamically displayed in UI

DATABASE DESIGN

Tables

Users – stores user accounts and roles
Properties – stores property details
Inquiries – stores user messages
Transactions – stores sales/rental data
Favorites – stores saved properties

Relationships

Properties linked to Users (agentId)
Favorites linked to Properties
Inquiries linked to Properties and Users
Transactions linked to Properties

Advanced Database Features

View

PropertyListingView

Combines Properties and Users
Displays agent name with property

Trigger

Automatically updates property status when a transaction occurs

Stored Procedures

AddOrUpdateUser
ProcessTransaction

TECHNOLOGIES USED

Frontend: HTML, CSS
Backend: PHP
Database: MySQL
Server: XAMPP
Tools: phpMyAdmin

IMPLEMENTATION DETAILS

PHP used to handle server-side logic and database interaction
Prepared statements used for secure queries
Sessions used to manage user authentication
Dynamic pages created using PHP loops and database queries
File upload used for property images
SQL queries used for filtering and data retrieval

CHALLENGES FACED

Handling foreign key constraints when deleting properties
Ensuring images display correctly from file paths
Debugging PHP session issues
Implementing secure password authentication
Integrating multiple features into a consistent UI

TESTING

The system was tested by:

Logging in as different user roles
Adding, editing, and deleting properties
Saving and removing favorites
Submitting inquiries
Searching properties by filters
Viewing property details

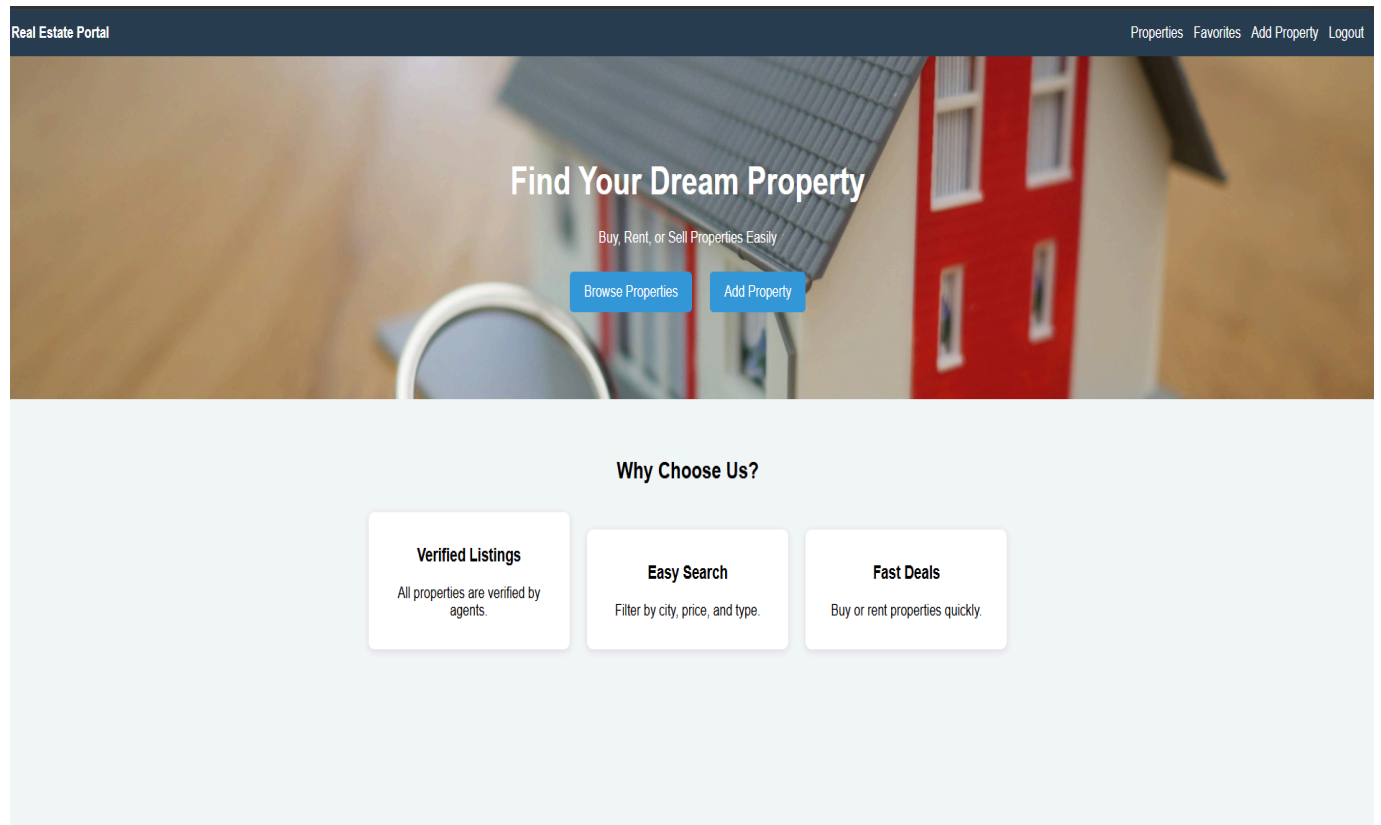
CONCLUSION

The Real Estate Portal System successfully fulfills all project requirements. It demonstrates the integration of database design, backend development, and frontend interface. The system provides a complete solution for managing and browsing real estate listings.

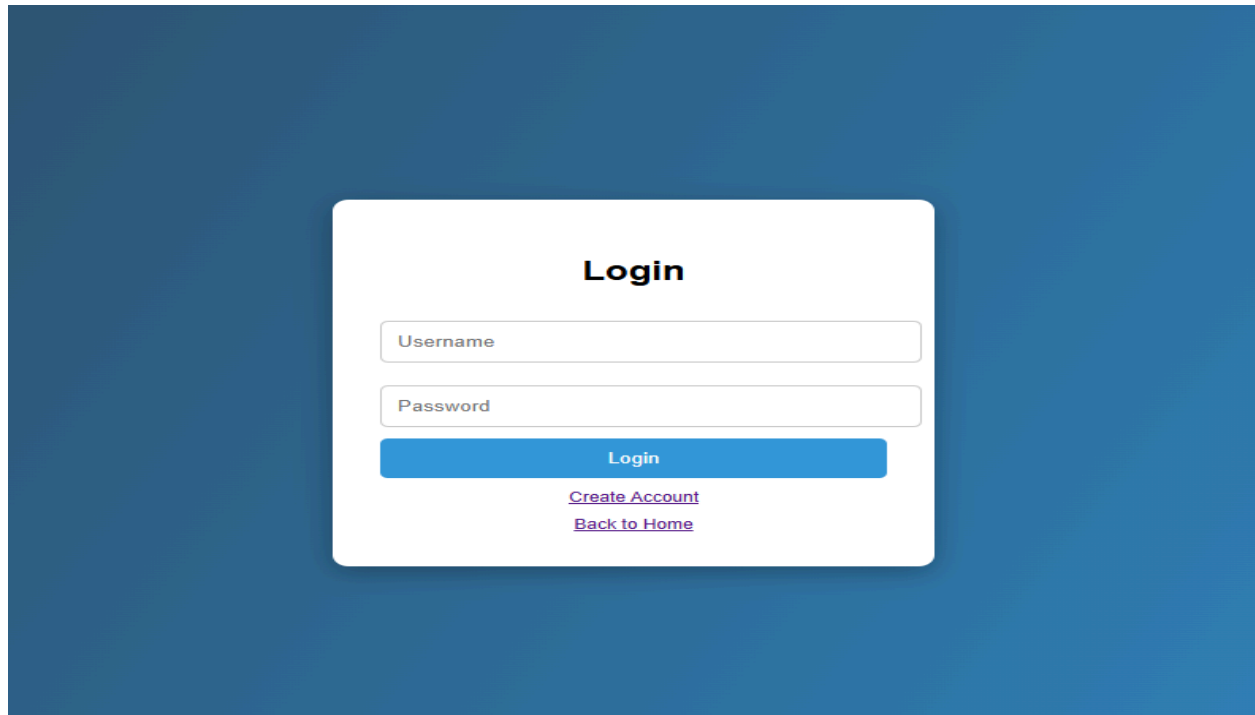
11. SCREENSHOTS (REQUIRED)

Insert screenshots of:

Home Page:



Login Page:

The image shows a login page with a dark blue background. In the center is a white rounded rectangle containing the login form. The form has a title 'Login', two input fields for 'Username' and 'Password', a blue 'Login' button, and two links: 'Create Account' and 'Back to Home'.

Login

Username

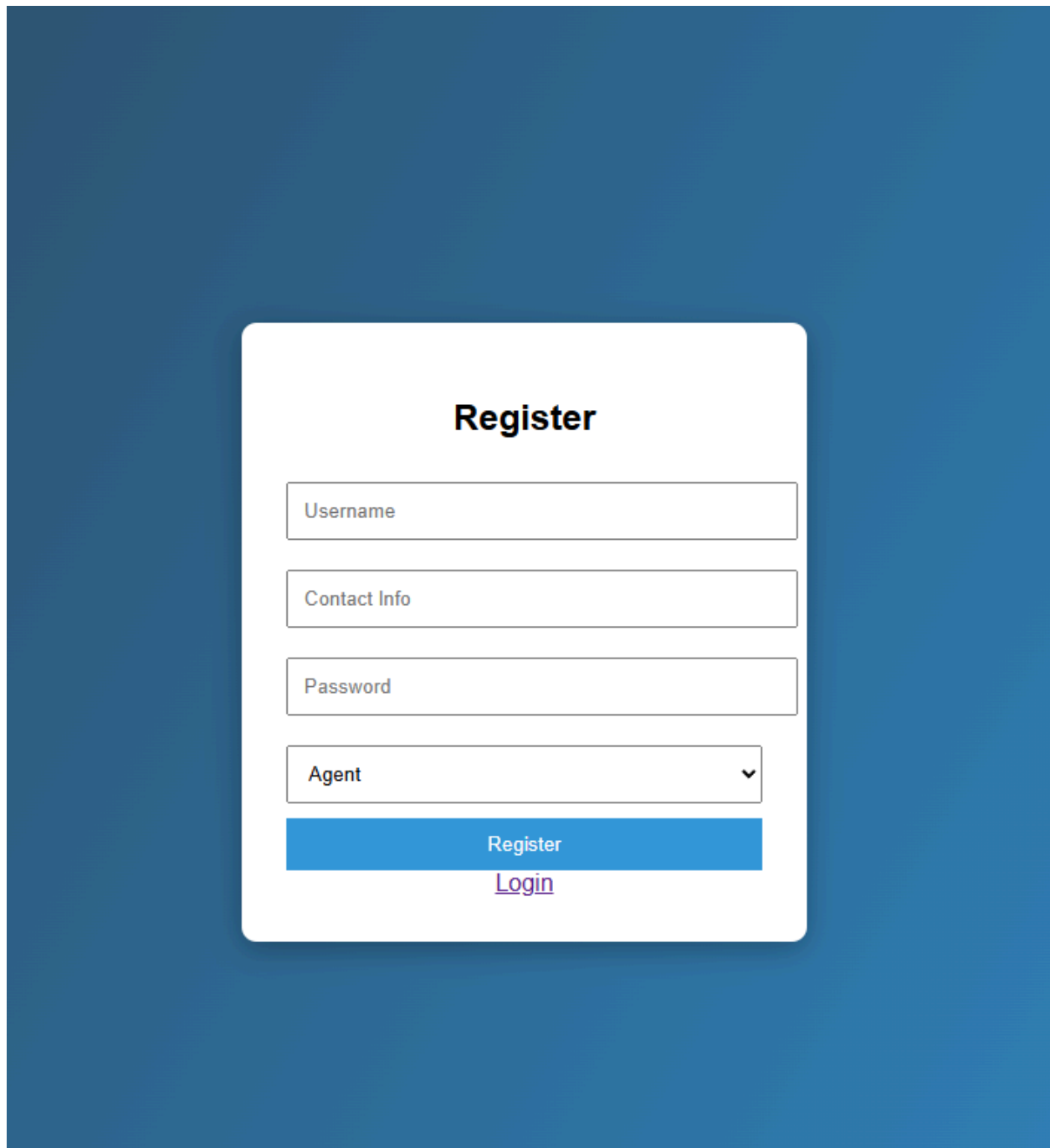
Password

Login

[Create Account](#)

[Back to Home](#)

Register Page:

A registration form is centered on a solid blue background. The form is a white rounded rectangle with a subtle drop shadow. At the top of the form is the title 'Register' in bold black text. Below the title are four input fields: 'Username', 'Contact Info', 'Password', and 'Agent'. The 'Agent' field is a dropdown menu with a downward arrow icon. Below these fields is a solid blue 'Register' button. Underneath the button is a purple, underlined link for 'Login'.

Register

Username

Contact Info

Password

Agent ▼

Register

[Login](#)

Properties Page

Available Properties

City

Max Price

Search



Modern Downtown Loft with Skyline View

New York

\$725000.00

available

Agent:

View Details

Save Favorite

Edit Delete



Coastal Family Home Near Boardwalk

Long Beach

\$890000.00

available

Agent:

View Details

Save Favorite

Edit Delete



Compact Studio Near University District

Newark

\$255000.00

available

Agent:

View Details

Save Favorite

Edit Delete



Renovated Brownstone with Private Backyard

Hoboken

\$975000.00

available

Agent:

View Details

Save Favorite

Edit Delete



Luxury High-Rise Condo with River Views

Jersey City

\$650000.00

available

Agent:

View Details

Save Favorite

Edit Delete

Property Details Page

Property Details



Coastal Family Home Near Boardwalk

\$890000.00

Type: House

City: Long Beach

Status: available

Agent:

Add Property Page

Add New Property

Property Information

Title

Example: Modern Downtown Loft

Property Type

Apartment, House, Condo...

Address

Street address

City

City

Price

Price

Status

Available

Property Image

Choose File

No file chosen

Add Property

[Back to Properties](#)

Favorites Page

My Favorite Properties

[Back to Properties](#)



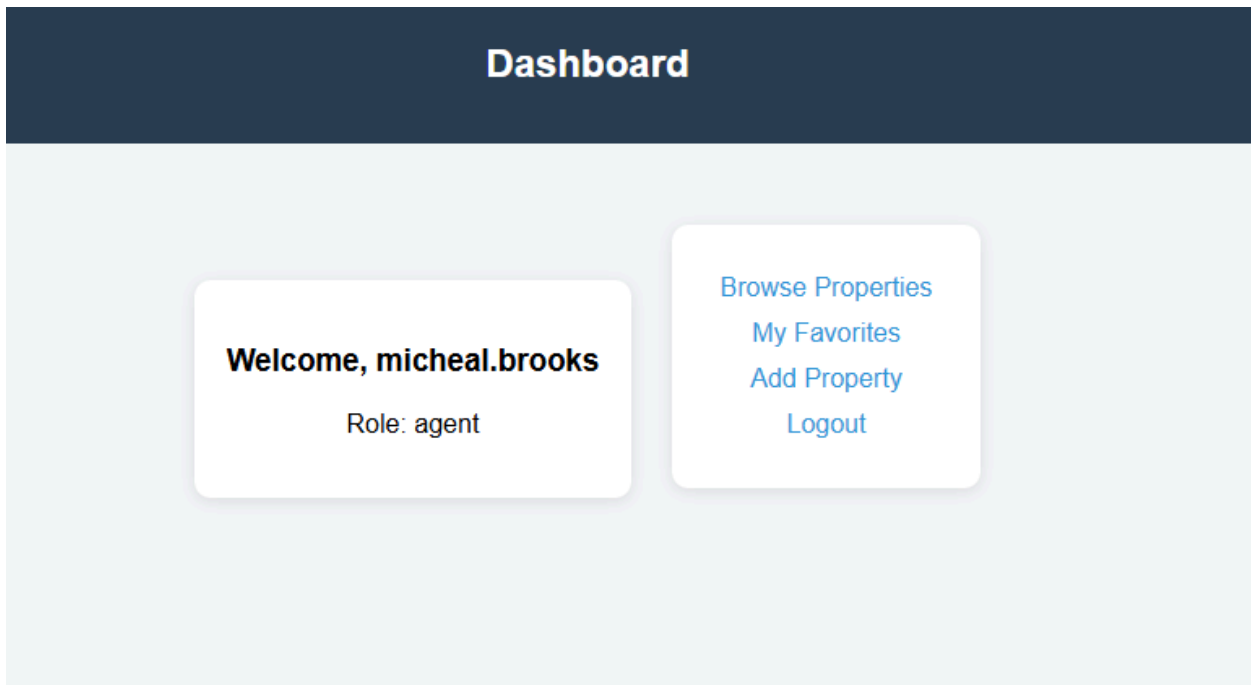
Coastal Family Home Near Boardwalk

Long Beach

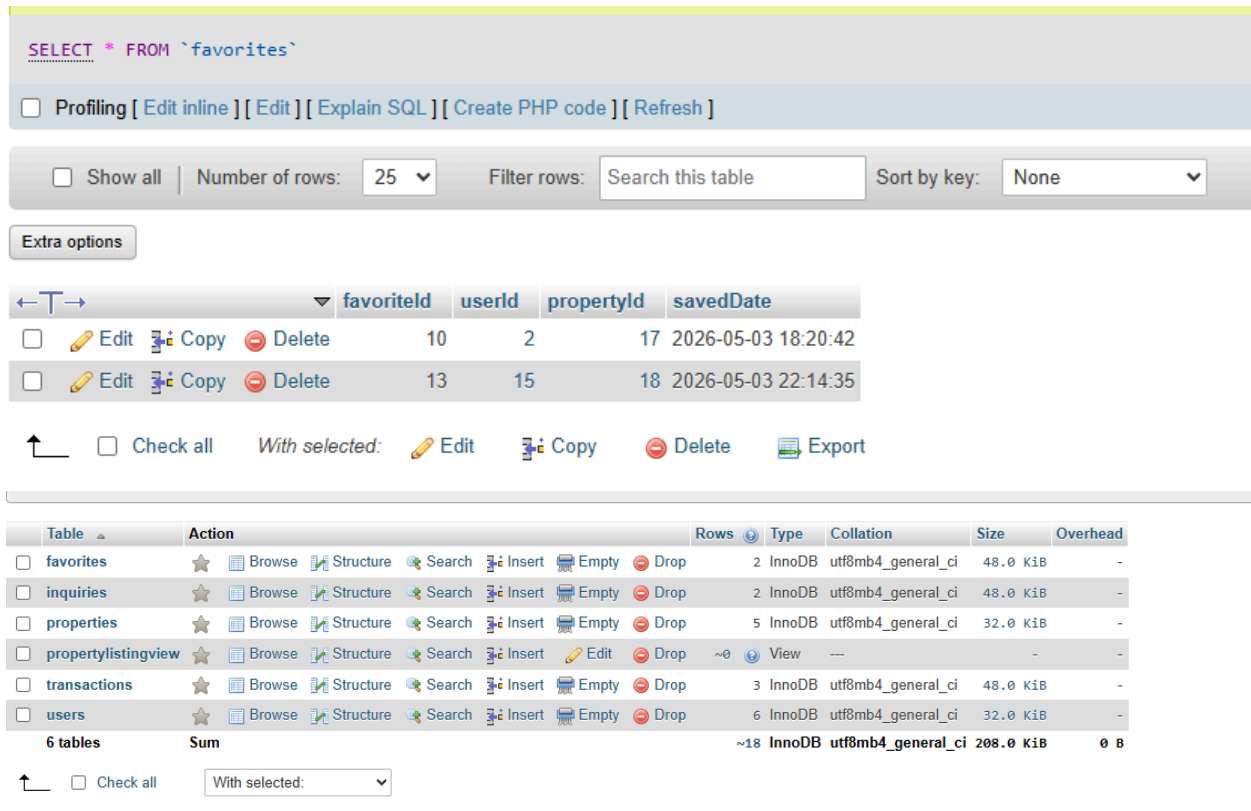
\$890000.00

[View Details](#)

Dashboard Page:



Database Tables (phpMyAdmin):



`SELECT * FROM `inquiries``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

					inquiryId	userId	propertyId	message	inquiryDate
☐					7	2	17	Interested	2026-05-03 18:43:55
☐					9	2	20	intersted	2026-05-03 18:52:20

☐ Check all | With selected: Edit Copy Delete Export

`SELECT * FROM `properties``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: | Filter rows:

You have a previously saved query. Click Get auto-saved query to load the query.

Extra options

					propertyId	title	propertyType	address	city	price	status	agentId	image
☐					17	Modern Downtown Loft with Skyline View	Apartment	88 Hudson St	New York	725000.00	sold	1	images/property1.webp
☐					18	Coastal Family Home Near Boardwalk	House	14 Seaside Drive	Long Beach	890000.00	rented	1	images/property2.jpg
☐					19	Compact Studio Near University District	Studio	221 College Ave	Newark	255000.00	sold	1	images/property3.jpg
☐					20	Renovated Brownstone with Private Backyard	House	45 Willow Ave	Hoboken	975000.00	available	1	images/property4.jpg
☐					21	Luxury High-Rise Condo with River Views	Condo	300 Riverfront Blvd, Apt 22C	Jersey City	650000.00	available	1	images/property5.jpg

☐ Check all | With selected: Edit Copy Delete Export

`SELECT * FROM `transactions``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: | Filter rows:

You have a previously saved query. Click Get auto-saved query to load the query.

Extra options

					transactionId	propertyId	userId	transactionType	transactionDate	amount
☐					16	17	4	sale	2026-05-03 22:57:33	710000.00
☐					17	18	1	rental	2026-05-03 22:57:33	3200.00
☐					18	19	4	sale	2026-05-03 22:57:33	248000.00

☐ Check all | With selected: Edit Copy Delete Export

`SELECT * FROM `users``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		userId	userName	contactInfo	passwordHash	userType
☐	Edit Copy Delete	1			\$2y\$10\$2s1pg5F20GjK04nflHv04oebuj5YLGlg71ETnNmy.W4R...	agent
☐	Edit Copy Delete	2	johna		\$2y\$10\$HryD8lkWOWVw.XUzdYzauEe4zq2ZXgPNBZnpypMcmH...	agent
☐	Edit Copy Delete	3	michael.brooks	michael.brooks@gmail.com	\$1y\$demo	agent
☐	Edit Copy Delete	4	sophia.reed	sophia.reed@yahoo.com	\$6y\$demo	buyer
☐	Edit Copy Delete	5	daniel.khan	dkhan@outlook.com	\$3y\$demo	renter
☐	Edit Copy Delete	15	micheal.brooks	1451235689	\$2y\$10\$TQbUqfQZRNuA96KOvFQCaOPkE/NolgBEOWus7aSvviD...	agent

[↶](#) ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

SQL View Output:

`SELECT * FROM `propertylistingview``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

You have a previously saved query. Click Get auto-saved query to load the query.

		propertyId	title	propertyType	city	price	status	image	agentName
☐	Edit Copy Delete	17	Modern Downtown Loft with Skyline View	Apartment	New York	725000.00	sold	images/property1.webp	
☐	Edit Copy Delete	18	Coastal Family Home Near Boardwalk	House	Long Beach	890000.00	rented	images/property2.jpg	
☐	Edit Copy Delete	19	Compact Studio Near University District	Studio	Newark	255000.00	sold	images/property3.jpg	
☐	Edit Copy Delete	20	Renovated Brownstone with Private Backyard	House	Hoboken	975000.00	available	images/property4.jpg	
☐	Edit Copy Delete	21	Luxury High-Rise Condo with River Views	Condo	Jersey City	650000.00	available	images/property5.jpg	

[↶](#) ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)